

# EE 371 Project: FM Signal Analysis and Demodulation

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## 1 Introduction and Assumptions

The objective of this project is to analyze and reconstruct a baseband FM signal recorded via SDR [1]. The process involves spectrum analysis, instantaneous frequency extraction, mono isolation, and de-emphasis.

**Assumption on Units:** Since the exact calibration constants of the SDR hardware and the transmitter's frequency sensitivity ( $k_f$ ) are not provided, standard unitary gain is assumed. Therefore, the received signal amplitude is presented in **Volts [V]**, and the demodulated signal amplitude corresponds directly to the **Frequency Deviation in [Hz]**. Absolute frequency deviation values are interpreted qualitatively rather than quantitatively.

## 2 Spectrum Analysis of the Received Signal

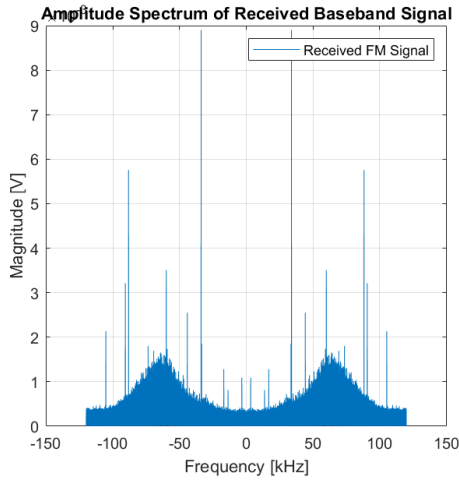
The dataset consists of a down-converted FM broadcast signal ( $f_s = 240$  kHz). The amplitude spectrum (FFT) in **Figure 1a** shows the signal energy concentrated around the carrier  $f_c = 60$  kHz.

## 3 FM Demodulation

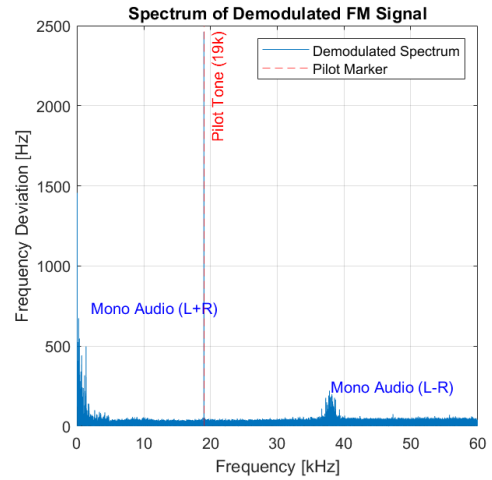
To extract the message in the digital domain, the **Hilbert Transform** was utilized instead of analog envelope detection [2]. The analytic signal  $z(t)$  yields the instantaneous phase  $\phi(t)$ , and its derivative gives the frequency:

$$f_i(t) = \frac{1}{2\pi} \frac{d\phi(t)}{dt} \quad (1)$$

Subtracting  $f_c = 60$  kHz reveals the baseband spectrum (**Figure 1b**). The plot clearly identifies the **Mono (L+R)** band (0-15 kHz), the **Pilot Tone** at 19 kHz, and the Stereo (L-R) components at higher frequencies.



(a) Spectrum of received signal [V]. Energy at  $f_c = 60$  kHz.



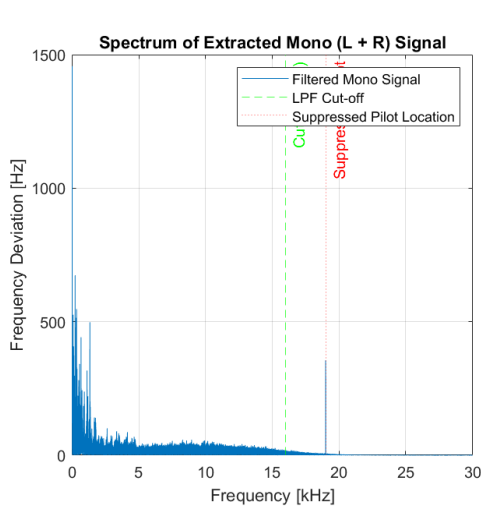
(b) Demodulated spectrum [Hz]. Pilot at 19 kHz is visible.

## 4 Mono Extraction and Filtering

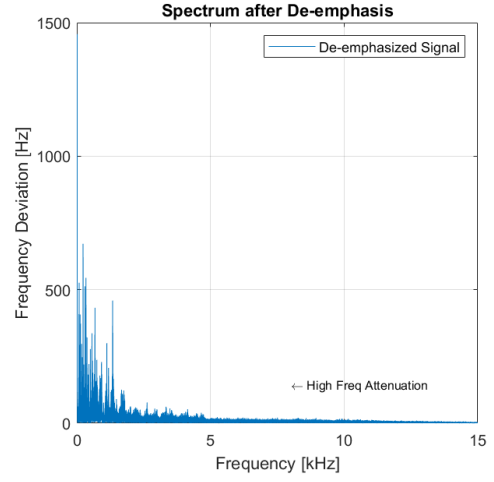
A 5th-order Butterworth **Low-Pass Filter** ( $f_{cut} = 16$  kHz) was designed to isolate the mono component and reject the pilot tone [2]. Zero-phase filtering was applied. As seen in **Figure 2a**, the pilot tone amplitude is significantly reduced. This residual component is expected due to the finite roll-off of the filter at 19 kHz, but it is sufficiently suppressed to prevent audio distortion.

## 5 De-emphasis

To compensate for pre-emphasis, a **De-emphasis filter** ( $\tau = 50\mu s$ ) was implemented as a first-order LPF ( $f_{corner} \approx 3183$  Hz). **Figure 2b** demonstrates the expected attenuation of high-frequency components, reducing noise.



(a) Mono signal spectrum [Hz]. Pilot is suppressed.



(b) Spectrum after De-emphasis [Hz]. High freq attenuated.

## 6 Final Reconstruction and Conclusion

The signal was decimated to  $f_{audio} = 48$  kHz. The final audio was normalized and saved as `recovered_audio.wav`. Listening tests confirmed clarity. Additionally, a bonus task involving FM modulation ( $f_c = 110$  kHz) was verified (`fav.mat`).

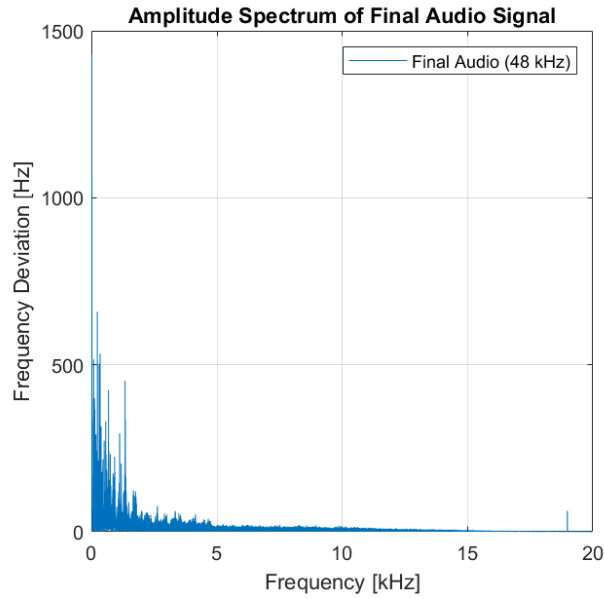


Figure 3: Final reconstructed audio spectrum ( $f_s = 48$  kHz).

## References

- [1] A. E. Pusane, “EE 371 Project Description,” Boğaziçi University, Fall 2025.
- [2] B. P. Lathi and Z. Ding, *Modern Digital and Analog Communication Systems*, 4th ed. Oxford Univ. Press, 2009.